



AI-Powered Ta'lim Platformasi

# MASTER PLATFORM IMPLEMENTATION PLAN

Version 1.0

|            |                                        |
|------------|----------------------------------------|
| Hujjat     | MASTER_PLATFORM_IMPLEMENTATION_PLAN_v1 |
| Sana       | 2026-06-25                             |
| Asoslangan | BLUEPRINT_v2 + AUDIT_v1                |
| Muallif    | CTO Office                             |
| Holat      | KONFIDENSIAL                           |

# 1. OLDINGI HUJJATLAR SHARHI

**Sharh maqsadi:** MASTER\_PLATFORM\_AUDIT\_v1 va MASTER\_PLATFORM\_BLUEPRINT\_v2 hujjatlarini CTO sifatida qayta ko'rib chiqdim. Quyida har bir hujjatning kuchli va zaif tomonlari, xato tavsiyalar va tuzatish kerak bo'lgan joylar keltirilgan.

## 1.1 Kuchli Tomonlar (Ikkala hujjatda)

- Platform modullari to'liq ro'yxatlangan (24 ta modul) va holati ko'rsatilgan
- Database ER diagrammasi — munosabatlar aniq
- Xavfsizlik muammolari obyektiv aniqlangan (role bug, email verify)
- AI tizim arxitektura diagrammasi mavjud
- TOP 100 yaxshilanishlar muhimlik bo'yicha tartiblangan
- Qurilmaydigan narsalar ro'yxati — muhim strategik qaror

## 1.2 Kamchiliklar va Xatolar

| Kamchilik                                                                                                 | Ta'sir                       |
|-----------------------------------------------------------------------------------------------------------|------------------------------|
| MVP aniq belgilanmagan — V1.0 uchun MAJBURIY narsalar ko'rsatilmagan                                      | <div><div></div>Kritik</div> |
| Qattiq amalga oshirish tartibi yo'q — TOP 100 bor, lekin qaysi biri BIRINCHI ekanligini aniq ko'rsatmaydi | <div><div></div>Kritik</div> |
| Texnik qoidalar yo'q — coding standards, naming conventions, folder structure                             | <div><div></div>Yuqori</div> |
| Dependency mapping yo'q — har bir vazifa nimaga bog'liq                                                   | <div><div></div>Yuqori</div> |
| Playwright tests "8-12 kun" — bu realistik emas, 1-2 oy kerak                                             | <div><div></div>Yuqori</div> |

## 1.3 Tushirib Qoldirilgan Muhim Muammolar

**TUSHIRILGAN: Supabase API Limit xavfi** — Free tier Edge Function 500K so'rov/oy. AI chat intensiv foydalanishda 1 hafta ichida limitga yetishi mumkin. PRO plan budget rejasi yo'q.

**TUSHIRILGAN: Gemini API xarajat baholash** — 100 faol talabada oylik xarajat \$50–200. Budget rejasi hujjatda yo'q.

**TUSHIRILGAN: sb\_publishable\_\* kalit formati** — Yangi Supabase kalit formati — ba'zi edge cases monitoring kerak.

## 2. O'ZGARTIRISH KERAK BO'LGAN TAVSIYALAR

| Oldingi Tavsiya                     | Yangi Tavsiya            | Sabab                                             |
|-------------------------------------|--------------------------|---------------------------------------------------|
| Zustand "3 oy ichida"               | V2.0 ga qoldirish        | Context API hozir yetarli. Premature optimization |
| TanStack Query "6 oy ichida"        | V2.0 ga qoldirish        | Supabase built-in methods yetarli                 |
| Feature-based refactor "9–14 hafta" | MVP chiqarilgandan keyin | MVP dan oldin refactor — vaqt isrof               |
| AI rate limiting "Upstash Redis"    | Avval Supabase DB ile    | Redis extra infra xarajati ortiqcha               |
| pgvector "Bosqich 3"                | V2.0 da, minimal 6 oy    | Infra murakkab — MVP uchun zarur emas             |
| White label MVP da                  | V3.0 da                  | Murakkab — MVP dan oldin vaqt isrof               |
| Mobile app V2.0 da                  | V3.0 ga ko'chirish       | 3–6 oy alohida loyiha — V2 ga juda erta           |

### 3. QILINMAYDIGAN TAVSIYALAR

| Tavsiya                         | Sabab                                          | Qayta ko'rish           |
|---------------------------------|------------------------------------------------|-------------------------|
| GraphQL API                     | REST ham yo'q. Supabase SDK yetarli            | Hech qachon             |
| Custom video conferencing       | Zoom/Meet API arzonroq va ishonchli            | V3.0 da 3rd party       |
| Blockchain sertifikatlar        | Ortiqcha murakkablik. DB verification yetarli  | Hech qachon             |
| Native iOS/Android              | Expo yetarli. Native — 3x xarajat              | Hech qachon             |
| Self-hosted AI (LLaMA)          | GPU — \$500+/oy. Gemini optimal                | Hech qachon             |
| Custom CMS                      | Supabase DB yetarli                            | Hech qachon             |
| Electron desktop app            | PWA yetarli. 150MB+ bundle                     | Hech qachon             |
| AI fine-tuning                  | Dataset + GPU team. Prompt engineering yetarli | Hech qachon (hozircha)  |
| Mikroservis arxitektura         | Monorepo + Edge Functions yetarli              | V4.0+ real scale bo'lsa |
| Real-time collaborative editing | OT algoritmlar — yillar                        | V4.0+ da Liveblocks     |

## 4. YAKUNIY ARXITEKTURA

### 4.1 O'zgarmaydigan Stack Qarorlari

| Qatlam   | Texnologiya                      | O'zgartirish qachon                     |
|----------|----------------------------------|-----------------------------------------|
| Frontend | React 19 + TypeScript            | Hech qachon (V1–V3)                     |
| Build    | Vite 8                           | Hech qachon                             |
| Styling  | Tailwind CSS v4                  | Faqat v5 chiqsa upgrade                 |
| Backend  | Supabase (Auth+DB+Storage+Edge)  | Faqat real scale kerak bo'lsa           |
| AI       | Gemini 2.5 Flash                 | AIProvider interface orqali yashirilgan |
| State    | React Context + useReducer       | V2.0 da Zustand qo'shimcha              |
| Testing  | Playwright (E2E) + Vitest (unit) | Vitest MVP da qo'shish                  |

### 4.2 Brand Konfiguratsiyasi (MVP da MAJBURIY)

```
// src/config/brand.ts — HAMMA NARSA BU YERDAN
export const BRAND = {
  platform: {
    name: import.meta.env.VITE_PLATFORM_NAME ?? "YordamchiAI",
    tagline: import.meta.env.VITE_PLATFORM_TAGLINE ?? "AI-powered Learning",
  },
  ai: {
    name: import.meta.env.VITE_AI_NAME ?? "AI Yordamchi",
    tagline: import.meta.env.VITE_AI_TAGLINE ?? "Powered by Gemini 2.5 Flash",
    avatarPath: import.meta.env.VITE_AI_AVATAR ?? "/ai-avatar.jpg",
    model: import.meta.env.VITE_AI_MODEL ?? "gemini-2.5-flash",
    maxTokens: parseInt(import.meta.env.VITE_AI_MAX_TOKENS ?? "1500"),
  },
  colors: {
    primary: import.meta.env.VITE_COLOR_PRIMARY ?? "#7c3aed",
    secondary: import.meta.env.VITE_COLOR_SECONDARY ?? "#4f46e5",
  },
} as const
```

## 5. MVP — VERSION 1.0 (Majburiy)

**MVP tamoyili:** Faqat platforma ISHLASHI uchun zarur bo'lgan narsalar. Har bir qo'shimcha xususiyat chiqarishni kechiktiradi.

### 5.1 Authentication

| Xususiyat                         | Sarlavha                |
|-----------------------------------|-------------------------|
| Role self-selection olib tashlash | <span>● BLOCKER</span>  |
| Email verification                | <span>● BLOCKER</span>  |
| /reset-password route             | <span>● BLOCKER</span>  |
| Parol o'zgartirish (in-app)       | <span>● Majburiy</span> |

### 5.2 Infrastructure

| Xususiyat                        | Sarlavha                |
|----------------------------------|-------------------------|
| ErrorBoundary barcha layoutlarda | <span>● BLOCKER</span>  |
| Toast notification system        | <span>● BLOCKER</span>  |
| Code splitting (React.lazy)      | <span>● BLOCKER</span>  |
| avatars Storage bucket           | <span>● BLOCKER</span>  |
| profiles.updated_at migration    | <span>● BLOCKER</span>  |
| brand.ts konfiguratsiya          | <span>● Majburiy</span> |

### 5.3 Identity

| Xususiyat                         | Sarlavha                |
|-----------------------------------|-------------------------|
| ProfileContext barcha sahifalarda | <span>● BLOCKER</span>  |
| Avatar upload (student profil)    | <span>● Majburiy</span> |
| UserAvatar unification            | <span>● Majburiy</span> |

### 5.4 AI System

| Xususiyat                   | Sarlavha               |
|-----------------------------|------------------------|
| Rate limiting (Supabase DB) | <span>● BLOCKER</span> |

|                                    |                                                                                                     |
|------------------------------------|-----------------------------------------------------------------------------------------------------|
| Token usage tracking               |  <b>BLOCKER</b>  |
| Haqiqiy streaming (ReadableStream) |  <b>Majburiy</b> |
| Context window sliding (20 xabar)  |  <b>Majburiy</b> |

## 5.5 MVP dan CHIQRILGANLAR (V2.0 ga)

- Homework moduli, Notifications, Leaderboard, Analytics, AI memory
- Voice input, File attachments, Comments, Announcements, Calendar
- Google OAuth, 2FA, White label, Payments, Marketplace, Mobile app

## 6. VERSION 2.0 — KEYIN

| Modul                | Xususiyatlar                              | Vaqt      |
|----------------------|-------------------------------------------|-----------|
| Homework             | DB, talaba topshirish, baholash, deadline | 3–4 hafta |
| Notifications        | Push, email, in-app notification center   | 2–3 hafta |
| Leaderboard          | Monthly/all-time, filters                 | 1–2 hafta |
| Analytics            | Admin dashboard, Teacher reports, charts  | 3–4 hafta |
| AI memory (pgvector) | Suhbatlar o'rtasida xotira                | 3–4 hafta |
| Video player         | Lesson progress, notes                    | 1–2 hafta |
| Google OAuth         | Tez kirish                                | 1 hafta   |
| Zustand (agar kerak) | State performance                         | 1 hafta   |



## 7. VERSION 3.0 — UZOQ MUDDATLI

| Modul                                  | Qiyinlik   |
|----------------------------------------|------------|
| Payments + Subscriptions + Marketplace | Katta      |
| Mobile App (Expo)                      | Katta      |
| AI Tutor Mode + AI Homework Checker    | Juda katta |
| White Label Complete                   | Katta      |
| Multi-tenant Architecture              | Juda katta |
| Public API                             | Katta      |
| Parent Portal + Community              | O'rta      |
| Live Classes (Zoom/Meet)               | O'rta      |

## 8. QATTIQ AMALGA OSHIRISH TARTIBI

**MUHIM QOIDA:** Bu tartibdan OG'ISHISH MUMKIN EMAS. Har bir sprint keyingisining asosi.

### Sprint 1 — Xavfsizlik (1–5 ish kuni) ● BLOCKER

| #   | Vazifa                            | Kun  | Risk   |
|-----|-----------------------------------|------|--------|
| 1.1 | Role self-selection olib tashlash | 0.5  | KRITIK |
| 1.2 | Email verification yoqish         | 0.25 | KRITIK |
| 1.3 | Keraksiz fayllarni o'chirish      | 0.1  | Past   |
| 1.4 | profiles.updated_at migration     | 0.25 | O'rta  |
| 1.5 | avatars bucket yaratish           | 0.25 | KRITIK |
| 1.6 | /reset-password sahifasi          | 1    | O'rta  |

### Sprint 2 — Barqarorlik (6–10 ish kuni)

| #   | Vazifa                                    | Kun | Risk   |
|-----|-------------------------------------------|-----|--------|
| 2.1 | ErrorBoundary barcha layoutlarga          | 1   | KRITIK |
| 2.2 | Toast notification system                 | 1.5 | O'rta  |
| 2.3 | ConfirmDialog shared component            | 1   | O'rta  |
| 2.4 | Code splitting: React.lazy()              | 2   | O'rta  |
| 2.5 | brand.ts config + hardcoded olib tashlash | 2   | Past   |

### Sprint 3 — Identity (11–18 ish kuni)

| #       | Vazifa                                                  | Kun |
|---------|---------------------------------------------------------|-----|
| 3.1–3.4 | ProfileContext → Student/Teacher/Admin layout va Navbar | 2   |
| 3.5     | Student ProfilePage → ProfileEditor + AvatarUploader    | 2   |
| 3.6     | Teacher ProfilePage → ProfileEditor                     | 1.5 |
| 3.7     | Avatar tizim unification (3 → 1)                        | 1.5 |

### Sprint 4 — AI System (19–26 ish kuni)

| # | Vazifa | Kun | Risk |
|---|--------|-----|------|
|---|--------|-----|------|

|     |                                           |   |                 |
|-----|-------------------------------------------|---|-----------------|
| 4.1 | AI rate limiting (Supabase DB)            | 2 | KRITIK (moliya) |
| 4.2 | Token usage tracking                      | 1 | KRITIK (moliya) |
| 4.3 | Context window sliding (20 xabar)         | 2 | O'rta           |
| 4.4 | Haqiqiy Gemini streaming (ReadableStream) | 5 | Yuqori          |
| 4.5 | AIAssistantPage: ChatContext ajratish     | 3 | O'rta           |

## Sprint 5 — LMS Core (27–40 ish kuni)

| #       | Vazifa                                                   | Kun |
|---------|----------------------------------------------------------|-----|
| 5.1–5.2 | Certificate DB + PDF + verification URL                  | 2.5 |
| 5.3     | Soft delete migrations                                   | 1   |
| 5.4–5.6 | Student: lesson progress, test flow, attendance calendar | 7   |
| 5.7     | Teacher: student progress view                           | 3   |
| 5.8     | Admin: user management yaxshilash                        | 2   |

## Sprint 6 — Sifat Nazorat (41–50 ish kuni)

| #       | Vazifa                                              | Kun |
|---------|-----------------------------------------------------|-----|
| 6.1–6.3 | Playwright E2E tests (login, AI chat, student flow) | 7   |
| 6.4     | Accessibility audit + fix (ARIA, keyboard)          | 3   |
| 6.5     | Mobile responsive audit + fix                       | 2   |

**V1.0 CHIQRISH MEZONI:** Sprint 1–6 barcha kritik vazifalar 100% tugallangan. E2E testlar o'tgan. Accessibility minimum. Initial load < 300KB. Error boundarylar barcha layoutda. Token tracking ishlaydi. Rate limiting ishlaydi.

## 9. TEXNIK QOIDALAR

### 9.1 Kodlash Standartlari

- TypeScript strict mode — **any ishlatis** TAQIQLANGAN
- ESLint + Prettier — barcha fayllar format qilingan
- Har bir async funksiya — try/catch + user-facing error xabari
- Har bir o'chirish — ConfirmDialog bilan tasdiqlash
- Commit: "feat: ...", "fix: ...", "chore: ..." formatida

### 9.2 Nomlash

| Element         | Format                      | Misol                               |
|-----------------|-----------------------------|-------------------------------------|
| React Component | PascalCase                  | UserAvatar, ProfileEditor           |
| Hook            | camelCase, "use" prefix     | useProfile, useToast                |
| Service         | camelCase, "Service" suffix | avatarService                       |
| Type/Interface  | PascalCase                  | UserProfile, ChatMessage            |
| Constant        | UPPER_SNAKE_CASE            | NEON_RING, AVATAR_SRC               |
| DB column       | snake_case                  | full_name, avatar_url               |
| File name       | kebab-case                  | user-avatar.tsx, ai-chat.service.ts |

### 9.3 Database Qoidalari

- Har yangi jadval: id (uuid), created\_at, updated\_at — MAJBURIY
- Har yangi jadval: RLS ENABLE + kamida 1 ta policy — SHART
- Muhim jadvallar: deleted\_at (nullable) — soft delete
- JSONB ustunlar: CHECK constraint yoki trigger bilan validatsiya
- Migration fayllar: 014\_... format, idempotent (IF NOT EXISTS)

### 9.4 UI Qoidalari

- Barcha icon-only button: aria-label="..." MAJBURIY
- Barcha form input: <label> yoki aria-label MAJBURIY
- Har bir sahifa: loading + error + empty state BO'LISHI SHART
- Rang: HECH QACHON hardcoded — faqat brand.ts va Tailwind
- Animatsiya: GPU transform only (translate, scale, opacity)
- Z-index: 10 (header), 20 (sidebar), 30 (modal), 40 (toast), 50 (tooltip)

## 9.5 AI Qoidalari

- Barcha AI so'rovlar: aiProvider.complete() orqali — to'g'ri chaqirish TAQIQLANGAN
- Har AI so'rov: rate limit tekshiruvi MAJBURIY (before call)
- Har AI javob: token count saqlash MAJBURIY (after call)
- AI avatar: faqat BRAND.ai.avatarPath dan — hardcoded yo'l TAQIQLANGAN
- Conversation tarix: max 20 xabar (unlimited TAQIQLANGAN)

## 9.6 Profil va Avatar Qoidalari

- AsomiddinAvatar — FAQAT AI kontentida (chat, thinking, header brand)
- UserAvatar — FAQAT foydalanuvchilar uchun
- Ikkala avatar bir-birini ALMASHTIRISHI MUMKIN EMAS
- Profil ma'lumotlari: faqat useProfile() orqali (useAuth().user direct ko'rsatish TAQIQLANGAN)

## 10. YAKUNIY QAROR

### Qurish kerak (MVP)

- Xavfsizlik sprint: role fix, email verify, password reset
- ErrorBoundary + Toast + Code splitting
- ProfileContext barcha sahifalarda
- AI rate limiting + token tracking
- Haqiqiy streaming + context window
- Certificate tizimi (DB + verification)
- E2E tests (critical flows)

### Qayta loyihalash kerak

- AIAssistantPage.tsx (700+ qator) → ChatContext + useChat hook ajratish
- Student/Teacher ProfilePage → ProfileEditor + AvatarUploader bilan
- Barcha layout footer → UserAvatar komponenti (hali inline div bor)
- settings jadval → brand.ts config bilan integratsiya

### Olib tashlash kerak

- MockAIProvider (export) — dead code
- try\_login.mjs, avatar-design.html — tasodifan commit
- Avvalgi gen\_\*.mjs scriptlar → docs/ ga ko'chirish

### Kutish kerak

- Homework, Notifications, AI memory → V2.0
- Payments, Mobile, White label → V3.0
- Zustand, TanStack Query → faqat kerak bo'lsa

**CTO XULOSASI:** Bu platforma kuchli poydevorga ega (Supabase + React 19 + Gemini). Eng katta muammo — o'lim o'rmoni: MVP chiqarilmagan, lekin ko'p xususiyatlar "qisman" tayyor. 6 ta BLOCKER hal qilinsa, V1.0 **2 oyda** chiqarilishi mumkin. Eng muhim qaror: FOCUS. Har bir qo'shimcha xususiyat MVP ni kechiktiradi. MVP ni chiqaring, foydalanuvchilar bilan sinovdan o'tkizing, keyin V2.0 quriladi.

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**MASTER PLATFORM IMPLEMENTATION PLAN v1.0**

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